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which is what the required equation becomes, on putting therein $d = 0$; the coefficients of the complete equation are seminvariants, and the terms in d are to be inserted by means of this property. The coefficient $3ac - b^2$ is reduced to zero by the operator

$$3a \partial_b + 2b \partial_c + c \partial_d;$$

it is therefore a seminvariant, and remains unaltered. The coefficient $c\sqrt{-\square}$ is what $\sqrt{-\square}$ becomes ($\square$ being the discriminant of the cubic equation) on putting therein $d = 0$, it is therefore to be changed into $\sqrt{-\square}$. Hence

4. For the cubic equation $(a, b, c, d \mid v, 1)^3$ the equation for $\theta (= \zeta^2(\alpha, \beta))$ is, $0 =$

$$(a^2, 0, 3ac - b^2, \sqrt{-\square})(\theta, 1)^3, \quad \text{where columns are headed } a^2X, \sqrt{-\square}X.$$

5. For the quartic equation $(a, b, c, d, e \mid v, 1)^4 = 0$;

$$\theta = \zeta^2(\alpha, \beta, \gamma) = -(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma).$$

The roots are

$$\begin{aligned} \theta_1 &= \zeta^4(\beta, \gamma, \delta), \\ \theta_2 &= -\zeta^4(\gamma, \delta, \alpha), \\ \theta_3 &= \zeta^4(\delta, \alpha, \beta), \\ \theta_4 &= -\zeta^4(\alpha, \beta, \gamma), \end{aligned}$$

the signs being in this case (and indeed for an equation of any even order) alternately positive and negative; in fact, if the equation is represented by $\varphi v = 0$, then the roots divided by $\zeta^2(\alpha, \beta, \gamma, \delta)$ should respectively be $\varphi'\alpha$, $\varphi'\beta$, $\varphi'\gamma$, $\varphi'\delta$, and this will be the case if the signs are taken as above.

6. Putting now $\delta = 0$ (and therefore $e = 0$) the roots become

$$\begin{aligned} \theta_1 &= \beta\gamma(\beta - \gamma), \\ \theta_2 &= \gamma\alpha(\gamma - \alpha), \\ \theta_3 &= \zeta^4(\delta, \alpha, \beta) \text{ (see text),} \\ \theta_4 &= -\zeta^4(\alpha, \beta, \gamma), \end{aligned}$$

where (α, β, γ) are the roots of $(a, b, c, d \mid v, 1)^3 = 0$. Let $\square$ denote the discriminant of the cubic function, then $\zeta^4(\alpha, \beta, \gamma) = -\frac{1}{a^2}\sqrt{-\square}$, and we have thus the linear equation; the cubic equation is

$$\Pi_1(\theta - \beta\gamma(\beta - \gamma)) = 0,$$

the coefficients of which can be calculated by the method of symmetric functions (see Annex No. 1).

7. The cubic equation being thus obtained, we have the two equations

$$\begin{aligned}\theta^3 a^2 + \theta^2 \cdot 0 + \theta(3ac - b^2) + c\sqrt{-\square} &= 0, \\ \theta^3 a^2 - \theta^2(3ac - b^2)d\sqrt{-\square}/\square + \dots &= 0,\end{aligned}$$

[the second factor being the cubic equation derived in Annex No. 1]; and multiplying these together, the resultant equation is, schematically,

$$\theta^6 \cdot a^4 + \theta^5 \cdot 0 + \theta^4 a^2(-9a^2d^2 + 4abcd - b^2d + \square) + \theta^3 \cdot a^2(-8a^2d + 4abc - b^3)d\sqrt{-\square} + \dots = 0,$$

where the coefficients have to be completed by adding the terms which contain e .

We have $\sqrt{-\square}$ in the place of $d\sqrt{-\square}$, and $\square$ in the place of $-d^2\square$. The coefficient $-8a^2d + 4abc - b^3$ is a seminvariant, and requires no alteration. The coefficient

$$-9a^2d^2 + 4abcd - b^3d + \square$$

that is,

$$\begin{aligned}&-9a^2d^2 + 4abcd - b^3d \\ &+ 27a^2d^2 - 18abcd + 4ac^3 + 4b^3d - b^2c^2;\end{aligned}$$

collects (with the contributions from $\square$) to the following table of monomials and integer coefficients:

a^2ce	-16
a^2d^2	+18
ab^2e	+6
$abcd$	-14
ac^3	+4
b^3d	+3
b^2c^2	-1

and the terms in e to be added to this, in order to make it a seminvariant, are easily found as displayed.

8. Hence, for the quartic equation $(a, b, c, d, e \mid v, 1)^4$, the equation for $\theta (= \zeta^2(\alpha, \beta, \gamma))$ is $0 =$

$$(a^6, e^2\sqrt{-\square}X, a^4X, a^2\sqrt{-\square}X, \square X)(\theta, 1)^4$$

where, expanding column by column,

column a^6 :	+1,
column $\sqrt{-\square}X$:	0,
column a^4X :	$a^2ce - 16, \quad a^2d^2 + 18, \quad ab^2e + 6,$ $abcd + 144, \quad ac^3 - 184, \quad b^4 - 7, \quad b^2c^2 + 3,$
column $a^2\sqrt{-\square}X$:	(the seminvariant $-8a^2d + 4abc - b^3$),
column $\square X$:	+1 · $\square$,

the whole giving the desired equation in θ for the quintic equation

$$(a, b, c, d, e, f \mid v, 1)^5 = 0;$$

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9. We have $\theta = \zeta^2(\alpha, \beta, \gamma, \delta)$, the roots being

$$\begin{aligned}\theta_1 &= \zeta^4(\beta, \gamma, \delta, \varepsilon), & \text{which for } \varepsilon = 0 \text{ becomes } \beta\gamma\delta\zeta^4(\beta, \gamma, \delta), \\ \theta_2 &= \zeta^4(\gamma, \delta, \varepsilon, \alpha), & -\gamma\delta\alpha\zeta^4(\gamma, \delta, \alpha), \\ \theta_3 &= \zeta^4(\delta, \varepsilon, \alpha, \beta), & \delta\alpha\beta\zeta^4(\delta, \alpha, \beta), \\ \theta_4 &= \zeta^4(\varepsilon, \alpha, \beta, \gamma), & -\alpha\beta\gamma\zeta^4(\alpha, \beta, \gamma), \\ \theta_5 &= \zeta^4(\alpha, \beta, \gamma, \delta), & \zeta^4(\alpha, \beta, \gamma, \delta).\end{aligned}$$

10. The linear equation is $\theta a^4 + e\sqrt{\square} = 0$; the quartic equation may be written $\Pi_1(\theta - \theta_i) = 0$, for the determination of which see Annex No. 2. The two equations are

$$\begin{aligned}\theta^4 \cdot a^6 + \theta^3 \cdot 0 + \theta^2 \cdot a^2(Me - \square) + \theta \cdot a^2 Ne\sqrt{\square} + e^2\square &= 0, \\ \theta \cdot a^4 + e\sqrt{\square} &= 0;\end{aligned}$$

and multiplying these together, the resulting equation is

$$\theta^5 \cdot a^{10} + \theta^4 \cdot 0 + \theta^3 \cdot a^6 (Me - \square) + \theta^2 \cdot a^4 (Ne + M) e\sqrt{\square} + \theta \cdot a^2 (N + a^2 e) e\sqrt{\square} \cdot (\cdot) + e^3 \square \sqrt{\square} = 0,$$

where the coefficients have to be completed by the addition of the terms in f . We have $\sqrt{\square}$ in the place of $e\sqrt{\square}$, and therefore $\square$ in the place of $e^2\square$.

11. The value of $Me - \square$ is

$$\begin{aligned}a^3e^3 &-160 \\ a^3bde &+132 \\ a^2ce^2 &+88 \\ a^2cd^2e &-177 \\ a^2d^4 &+27 \\ ab^2ce^2 &-97 \\ ab^2d^2e &+6 \\ abc^2de &+62 \\ abcd^3 &-18 \\ ac^4e &-12 \\ ac^3d^2 &+4 \\ b^4e^2 &+18 \\ b^3cde &-14 \\ b^3d^3 &+4 \\ b^2c^2e &+3 \\ b^2cd^2 &-1\end{aligned}$$

and the terms in f are found to be

$$\begin{aligned}a^3def &+300 \\ a^3bcef &-130 \\ a^3bd^2f &-120 \\ a^2c^2df &+40 \\ ab^2ef &+28 \\ ab^2cdf &+66 \\ abc^3f &-24 \\ b^4df &-16 \\ b^2c^2f &+6 \\ a^3cf^2 &-125 \\ a^2b^2f^2 &+50\end{aligned}$$

12. The value of $Ne + M$ is

$$\begin{array}{rcl}
 a^3e^3 & + & 80 \\
 a^2bde & - & 54 \\
 a^2ce^2 & - & 36 \\
 a^2cd^2 & + & 27 \\
 ab^2ce & + & 42 \\
 abc^2d & - & 18 \\
 ac^4 & + & 4 \\
 b^4e & - & 8 \\
 b^3cd & + & 4 \\
 b^2c^3 & - & 1
 \end{array}$$

and the terms in f to be added thereto are found to be

$$\begin{array}{rcl}
 a^3df & + & 50 \\
 a^2bcf & + & 30 \\
 ab^3f & - & 8
 \end{array}$$

13. The value of $N + a^2e$ is

$$\begin{array}{rcl}
 a^3e & - & 15 \\
 a^2bd & + & 6 \\
 a^2c^2 & + & 4 \\
 ab^2c & - & 5 \\
 b^4 & + & 1
 \end{array}$$

which is a seminvariant, and requires no addition.

14. Hence, for the quintic equation

$$(a, b, c, d, e, f \mid v, 1)^5 = 0,$$

the equation for $\theta (= \zeta^2(\alpha, \beta, \gamma, \delta))$ is, $0 =$

$$(a^{12}X, 0, a^6X, a^3\sqrt{\square}X, \square X, \square\sqrt{\square}X)(\theta, 1)^5,$$

with column entries

a^6X		$a^3\sqrt{\square}X$		$\square X$	
a^3cf^2	-125	a^3df	-50	a^3e	-15
a^3def	+300	a^3e^2	+80	a^2bd	+6
a^3e^3	-160	a^2bcf	+30	a^2c^2	+4
$a^2b^2f^2$	+50	a^2bde	-54	ab^2c	-5
a^2bde^2	+132	a^2ce^2	-36	b^4	+1
a^2bcef	-130	a^2cd^2	+27		
a^2bd^2f	-120	ab^3f	-8		
a^2c^2df	+40	ab^2ce	+42		
$a^2c^2e^2$	+88	abc^2d	-18		
a^2cd^3	-117	ac^4	+4		
a^2d^4	+27	b^4e	-8		
ab^3ef	+28	b^3cd	+4		
ab^2ce^2	-97	b^2c^3	-1		
ab^2d^2e	+6				
abc^2de	+62				
ab^2cdf	+66				
$abcd^3$	-18				
abc^3f	-24				
ac^4e	-12				
ac^3d^2	+4				
b^4df	-16				
b^4e^2	+18				
b^3cde	-14				
b^3d^3	+4				
b^2c^2f	+6				
b^2c^3e	+3				
b^2cd^2	-1				

and the columns headed $a^{12}X$, 0, and $\square\sqrt{\square}X$ have entries +1, 0, and +1, respectively.

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15. As a verification of this result, I remark that, taking for the quintic equation $v^5 + v^4 + v^3 + v^2 + v + 1 = 0$, the roots of this equation are $-1, \omega, \omega^2, -\omega, -\omega^2$, where ω is an imaginary cube root of unity ($\omega^2 + \omega + 1 = 0$). We ought to have $\zeta^4(\alpha, \beta, \gamma, \delta, \varepsilon) = -\sqrt{\square} = -36$; and this will be the case if, for instance, $\alpha, \beta, \gamma, \delta, \varepsilon$ are respectively $-1, \omega, \omega^2, -\omega^2, -\omega$. We have then

$$\zeta^4(\alpha, \beta, \gamma, \delta) = -1 - \omega \cdot -1 - \omega^2 \cdot -1 + \omega^2 \cdot \omega - \omega^2 \cdot \omega + \omega^2 \cdot 2\omega^2 = 6,$$

$$\zeta^4(\beta, \gamma, \delta, \varepsilon) = \omega - \omega^2 \cdot \omega + \omega^2 \cdot 2\omega \cdot 2\omega^2 \cdot \omega^2 + \omega \cdot -\omega^2 + \omega = -12,$$

$$\zeta^4(\gamma, \delta, \varepsilon, \alpha) = 2\omega^2 \cdot \omega + \omega \cdot \omega^2 + 1 \cdot -\omega^2 - \omega \cdot -\omega^2 + 1 \cdot -\omega + 1 = +6(\omega - \omega^2),$$

$$\zeta^4(\delta, \varepsilon, \alpha, \beta) = -\omega^2 + \omega \cdot -\omega^2 + 1 \cdot -\omega^2 - \omega \cdot -\omega + 1 \cdot -2\omega \cdot -1 - \omega = -6(\omega - \omega^2),$$

$$\zeta^4(\varepsilon, \alpha, \beta, \gamma) = -\omega + 1 \cdot -2\omega \cdot -\omega - \omega^2 \cdot -1 - \omega \cdot -1 - \omega \cdot \omega - \omega^2 = 6.$$

The equation in θ is thus $(\theta - 6)^2(\theta + 12)(\theta^2 + 108) = 0$, or multiplying out it is

$$(1, 0, 0, +432, -11664, +46656 \mid \theta, 1)^5 = 0,$$

which in fact (observing that $\sqrt{\square} = 36$) is what the preceding formula becomes for the equation $(1, 1, 1, 1, 1, 1 \mid v, 1)^5 = 0$.

The analogous verifications for the cubic and the quartic equations are as follows:

16. For the cubic, if the assumed equation is $v^3 + v^2 + v + 1 = 0$, the roots whereof are $-1, i, -i$ ($i^2 = -1$), then we should have $\zeta^4(\alpha, \beta, \gamma) = \sqrt{-\square} = 4i$, which will be the case if $\alpha, \beta, \gamma = -1, i, -i$ respectively, and the roots $\beta - \gamma, \gamma - \alpha, \alpha - \beta$ of the equation in θ then are $2i, -i + 1, -i - 1$, so that the equation in θ is $(\theta^2 + 2i\theta - 2)(\theta - 2i) = 0$, or

$$(1, 0, 2, 4i \mid \theta, 1)^3 = 0,$$

which (observing that $\sqrt{-\square} = 4i$) is what the formula for the equation in θ becomes for the equation $(1, 1, 1, 1 \mid v, 1)^3 = 0$.

17. For the quartic equation, taking this to be $v^4 + v^3 + v^2 + v + 1 = 0$, the roots are $\omega, \omega^2, \omega^3, \omega^4$, where ω is an imaginary fifth root of unity ($\omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0$), and putting $\alpha, \beta, \gamma, \delta$ equal to $\omega, \omega^2, \omega^3, \omega^4$ respectively, we have

$$-\sqrt{\square} = \zeta^4(\alpha, \beta, \gamma, \delta) = -5(\omega + \omega^4 - \omega^2 - \omega^3),$$

giving, as it should do, $\square = 125$. The equation in θ is therefore by the formula

$$(1, 0, 0, -25(\omega + \omega^4 - \omega^2 - \omega^3), 125 \mid \theta, 1)^4 = 0.$$

But the roots are

$$\begin{aligned}\theta_1 &= \zeta^4(\beta, \gamma, \delta) = -\omega^2 - \omega^3 \cdot \omega^2 - \omega^4 \cdot \omega^3 - \omega^4 = 2 - \omega + \omega^2 - 2\omega^3 = 2 - X, \\ \theta_2 &= -\zeta^4(\gamma, \delta, \alpha) = -\omega^3 - \omega^4 \cdot \omega^3 - \omega \cdot \omega^4 - \omega = -1 + 3\omega + 2\omega^2 + \omega^3 = -1 + Y, \\ \theta_3 &= \zeta^4(\delta, \alpha, \beta) = \omega^4 - \omega \cdot \omega^4 - \omega^2 \cdot \omega - \omega^2 = -4 - 3\omega - 2\omega^2 - \omega^3 = -4 - Y, \\ \theta_4 &= -\zeta^4(\alpha, \beta, \gamma) = -\omega - \omega^2 \cdot \omega - \omega^3 \cdot \omega^2 - \omega^3 = 3 + \omega - \omega^2 + 2\omega^3 = 3 + X,\end{aligned}$$

if, for shortness,

$$X = \omega - \omega^2 + 2\omega^3, \quad Y = 3\omega + 2\omega^2 + \omega^3.$$

The equation in θ is therefore

$$(\theta - 2 + X)(\theta + 1 - Y)(\theta + 4 + Y)(\theta - 3 - X) = 0,$$

where the left-hand side is the product of the factors, and

$$\begin{aligned}(\theta - 2 + X)(\theta - 3 - X) &= \theta^2 - 5\theta + 6 - X - X^2 = \theta^2 - 5\theta + 10 - 5(\omega + \omega^4), \\ (\theta + 1 - Y)(\theta + 4 + Y) &= \theta^2 + 5\theta + 4 - 3Y - Y^2 = \theta^2 + 5\theta + 10 - 5(\omega^2 + \omega^3),\end{aligned}$$

and the equation in θ is, therefore, as it should be,

$$(1, 0, 0, -25(\omega + \omega^4 - \omega^2 - \omega^3), 125 \mid \theta, 1)^4 = 0.$$

Passing from the denumerate to the standard forms:

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18. For the cubic equation $(a, b, c, d \mid v, 1)^3 = 0$, the equation for $\theta (= \zeta^2(\alpha, \beta))$ is $0 =$

$$(a^2X, 0, 9HX, \sqrt{-27\Box}X)(\theta, 1)^3,$$

with column entries

$$\begin{array}{ll} a^4X : & +1, \\ \text{column 2} : & 0, \\ 9HX : & ba^2c + 1, \\ \sqrt{-27\Box}X : & +1. \end{array}$$

19. For the quartic equation $(a, b, c, d, e \mid v, 1)^4 = 0$, the equation for $\theta (= \zeta^2(\alpha, \beta, \gamma))$ is $0 =$

$$(a^6X, 0, 96a^2X, 512\sqrt{\Box}X, 256\Box X)(\theta, 1)^4,$$

with the entries (collected column by column) including

$$\begin{array}{ll} 96a^4HX \text{ column} & \\ a^2ce & -1 \\ abc & +3 \\ b^2 & -2 \\ 512\sqrt{\Box}X \text{ column} & \\ a^2d^2 & +3 \\ ab^2e & +1 \\ abcd & +149 \\ b^4d & +8 \\ b^2c^2 & -6 \end{array}$$

and a final column $+1 \cdot \Box$ in $256\Box X$.

20. For the quintic equation $(a, b, c, d, e, f \mid v, 1)^5 = 0$, the equation for $\theta (= \zeta^2(\alpha, \beta, \gamma, \delta))$ is, $0 =$

$$(a^{12}X, 0, 625a^6X, 12500\sqrt{\Box}a^3X, 15625\Box X, 76125\Box\sqrt{\Box})(\theta, 1)^5,$$

[The long coefficient table printed below this display is not yet fully collated in this slice; the coefficient header above has been repaired from the source scan.]

21. I remark, with respect to the equation in θ , for the cubic, that it leads at once to the equation of differences. In fact we have

$$a^2\theta^3 + 9(ac - b^2)\theta + \sqrt{-27\Box} = \Pi_1\{\theta - (\alpha - \beta)\},$$

whence changing the sign of d ,

$$a^2\theta^3 + 9(ac - b^2)\theta - \sqrt{-27\Box} = \Pi_1\{\theta + (\alpha - \beta)\};$$

or multiplying the two equations and putting u for θ^2 ,

$$u\{a^2u + 9(ac - b^2)\}^2 + 27\Box = \Pi_1\{u - (\alpha - \beta)^2\},$$

that is, the equation of differences is

$$a^4u^3 + 18(ac - b^2)a^2u^2 + 81(ac - b^2)^2u + 27\Box = 0;$$

but this mode of composition is peculiar to the case of the cubic.

If in the several equations in θ we substitute for the seminvariants the covariants to which they respectively belong, we obtain as follows:

22. For the cubic equation $(a, b, c, d \mid x, y)^3 = 0$, the equation for

$$\theta [= (\beta - \gamma)^2(x - \alpha y)^2]$$

is, $0 =$

$$(U^3, 0, 9H, \sqrt{-27\Box}X)(\theta, 1)^3 = 0.$$

23. For the quartic equation $(a, b, c, d, e \mid x, y)^4 = 0$, the equation for

$$\theta [= (\beta - \gamma)(\gamma - \delta)(\delta - \beta)(x - \alpha y)^2]$$

is

$$(U^6, 0, 96U^3(-3JU + 21H), 512\sqrt{\Box}(-\Phi), 256\Box)(\theta, 1)^4 = 0.$$

24. And for the quintic equation $(a, b, c, d, e, f \mid x, y)^5 = 0$, denoting the covariants of the quintic as in 141, A the quintic itself, &c., and completing the expression for the coefficient of ϑ^3 , the equation for

$$\vartheta [= (\beta - \gamma)(\beta - \delta)(\beta - \varepsilon)(\gamma - \delta)(\gamma - \varepsilon)(\delta - \varepsilon)(x - \alpha y)^3]$$

is

$$\left\{ \begin{array}{l} A^{12}, \\ 0, \\ 625 A^6(48AJ - 80B^3 + 120BH - 50CG), \\ 12500\sqrt{\Box} A^3(4AB^3 - AH - 50CD), \\ 15625\Box(3A^2B + 25C^2), \\ 76125\Box\sqrt{\Box} \end{array} \right\} (\vartheta, 1)^5 = 0,$$

where the covariant which enters into the coefficient of ϑ^3 , being of the sixth degree in the coefficients, is not given in the Tables.

Its value (completed for me, from the first term, by Mr Davis) is

[a large table of monomial covariants of degree 6, occupying the body of

printed page 287, with monomials such as

a^4c^4ef	-2
$a^2c^4d^2$	+18
$a^2b^3c^4$	-30
$a^3b^4e^2$	+3100
$a^3b^4e^3$	-30
a^4b^4c	+18
a^5c^3d	+2
a^4b^4f	-32
$a^2b^2d^2$	+2
a^4bcd^2	+624
a^2bcde^2	-540
ab^3cd^2e	+1196
abc^2d^3	-840
ab^3cd^3	+1080
ab^4cd^3	+672240

and so on,

...

[the full table being too dense to set in line;

the reader is referred to the original printed page 287].

[viz. this is $= 48 AJ - 80 HI + 120 BI - 50 CO$ as above].

In the following two Annexes, the notation of the symmetric functions is the same as in my "Memoir on the Symmetric Functions of the Roots of an Equation," *Phil. Trans.*, vol. CXLVII. (1857), [147] and the values of the symmetric functions are taken from that memoir, the powers of a being restored by the principle of homogeneity. The suffixes of the Σ indicate the number of terms in the sum; thus in the first Annex

$$\Sigma_3 \beta \gamma (\beta - \gamma) (\gamma - \alpha) = \Sigma_3 (\beta \gamma^2 - \alpha \beta \gamma - \gamma^3 + \alpha \gamma^2);$$

the terms $\Sigma_3 (\beta \gamma + \alpha \beta)$ are equal to $\Sigma_3 \Sigma \beta$, the complete symmetric function; the correct result will be obtained (though of course neither of these equations is true) by writing $\Sigma_3 \beta \gamma = \frac{1}{2} \Sigma_3 \alpha^2 \beta$, $\Sigma_3 \alpha \gamma^2 = \frac{1}{2} \Sigma_3 \alpha^2 \beta$, and so in similar cases; the insertion of the suffix to the Σ very much facilitates the calculation, and is a check on its accuracy.

Annex No. 1, containing the calculation of the equation $\Pi_1(\theta - \theta_i) = 0$, where

$$\theta_1 = \beta \gamma (\beta - \gamma), \quad \theta_2 = \gamma \alpha (\gamma - \alpha), \quad \theta_3 = \alpha \beta (\alpha - \beta),$$

α, β, γ being the roots of the cubic equation $(a, b, c, d \mid v, 1)^3 = 0$.

We have

$$\Sigma \theta_i = \Sigma \beta \gamma (\beta - \gamma) = -(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) = \zeta^4(\alpha, \beta, \gamma) = a^{-2} \sqrt{-\square},$$

where $\square = 27a^2d^2 + \&c.$ is the discriminant of the cubic.

$$\Sigma\theta_1\theta_2 = \Sigma_3 \beta\gamma(\beta - \gamma) \cdot \gamma\alpha(\gamma - \alpha) = \alpha\beta\gamma \Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha),$$

where $\alpha\beta\gamma = -a^{-1}d$ and

$$\begin{aligned} \Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha) &= \Sigma_3 (\beta\gamma^2 - \alpha\beta\gamma - \gamma^3 + \alpha\gamma^2) \\ &= 3\Sigma_3 \alpha^2\beta - 3(\alpha\beta\gamma) - \Sigma_3 \alpha^3 \\ &= 3(21) - 3(13) - (3), \end{aligned}$$

which, expressed in terms of the power-sums, gives

$$\Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha) = a^{-3} (9a^2d - 4abc + b^3) \text{ (after reduction),}$$

so that

$$\Sigma\theta_1\theta_2 = -a^{-4} (9a^2d^2 - 4abcd + b^3d).$$

And lastly,

$$\begin{aligned} \theta_1\theta_2\theta_3 &= \alpha^2\beta^2\gamma^2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \\ &= -\alpha^2\beta^2\gamma^2\zeta^4(\alpha, \beta, \gamma) \\ &= -a^{-4}d^2\sqrt{-\square}, \end{aligned}$$

so that the equation is the one given above, No. 7.

Annex No. 2, containing the calculation of the equation $\Pi_1(\theta - \theta_i) = 0$, where

$$\theta_1 = \beta\gamma\delta\zeta^4(\beta, \gamma, \delta), \quad \theta_2 = -\gamma\delta\alpha\zeta^4(\gamma, \delta, \alpha), \quad \theta_3 = \delta\alpha\beta\zeta^4(\delta, \alpha, \beta), \quad \theta_4 = -\alpha\beta\gamma\zeta^4(\alpha, \beta, \gamma),$$

$\alpha, \beta, \gamma, \delta$ being the roots of the quartic equation $(a, b, c, d, e \mid v, 1)^4 = 0$.

$$\begin{aligned} \Sigma\theta_i &= (\alpha\beta\gamma\delta) \Sigma_4 \beta\gamma\delta\zeta^4(\beta, \gamma, \delta) \\ &= -(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta) \\ &= -\zeta^4(\alpha, \beta, \gamma, \delta), \end{aligned}$$

where $\square = 256a^3e^3 + \&c.$ is the discriminant of the quartic.

$$\begin{aligned} \Sigma\theta_1\theta_2 &= \Sigma_4 \theta_i\theta_j = \Sigma_4 \beta\gamma\alpha\zeta^4(\delta, \alpha, \beta) \cdot (-\alpha\beta\gamma)\zeta^4(\alpha, \beta, \gamma) \\ &= \Sigma_4 \beta\gamma(\delta - \alpha)(\delta - \beta)(\alpha - \beta) \cdot -\alpha\beta\gamma(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma) \\ &= -\alpha\beta\gamma\delta \Sigma_4 \alpha\beta(\alpha - \beta)^2(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta), \end{aligned}$$

where $\alpha\beta\gamma\delta = -a^{-1}e$, and*

* The signs of $\theta_1, \theta_2, \theta_3, \theta_4$ are taken account of implicitly.

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(continuing)

$$= \Sigma_4 \alpha \beta (\alpha - \beta)^2 \{ \alpha^2 \beta^2 - \alpha \beta (\alpha + \beta)(\gamma + \delta) + (\alpha^2 + \beta^2) \gamma \delta \\ + \alpha \beta (\gamma + \delta)^2 - \gamma \delta (\alpha + \beta)(\gamma + \delta) + \gamma^2 \delta^2 \}$$

viz., expanding term by term,

$$\begin{aligned} & \Sigma_4 \alpha^3 \beta^3 (\alpha - \beta)^2, \\ & - \Sigma_4 \alpha^2 \beta^2 (\alpha - \beta)^2 (\alpha + \beta) (\gamma + \delta), \\ & + \alpha \beta \gamma \delta \Sigma_4 (\alpha - \beta)^2 (\alpha^2 + \beta^2), \\ & + \Sigma_4 \alpha^2 \beta^2 (\alpha - \beta)^2 (\gamma + \delta)^2, \\ & - \alpha \beta \gamma \delta \Sigma_4 (\alpha - \beta)^2 (\alpha + \beta)(\gamma + \delta), \\ & + \alpha^2 \beta^2 \gamma^2 \delta^2 \Sigma_4 (\alpha - \beta)^2, \end{aligned}$$

which, expressed in the notation of the memoir [147] (the suffixes indicating term-counts), reduce respectively to the elementary symmetric functions

$$\begin{aligned} & \Sigma_2 \alpha^4 \beta^4 \quad (15), \\ & - \Sigma_3 \alpha^3 \beta^3 \gamma \quad (53), \\ & + \Sigma_4 \alpha^2 \beta^2 \gamma \quad (431), \\ & + \Sigma_5 \alpha^2 \beta^2 \gamma^2 \delta \quad (322), \\ & - \Sigma_6 \alpha^2 \beta \gamma \delta \quad (3211), \\ & + \Sigma_7 \alpha \beta \gamma \delta \quad (21^3), \end{aligned}$$

where for a moment a is put equal to unity. The value of the last-mentioned expression is then calculated as follows:

[A long column of intermediate sums is set down in the original; the principal contributing monomials, with their integer multipliers, are

$$\begin{array}{ll} bde & -4 - 12, \\ b^4e & +1 + 16, \\ b^2ce & +8 + 8, \\ b^2c^2e^2 & +73 - 8, \\ cb^4ce^2 & +73, \\ b^2c^2de & +311, \\ b^3c^2e^2d & +48, \\ \&c., & \&c.; \end{array}$$

after collecting, the final result equals

$$\Sigma_4 \alpha \beta (\alpha - \beta)^2 (\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = a^{-3} M e,$$

say.]

and restoring the powers of a by the principle of homogeneity, and putting

$$\begin{array}{rcl} a^3 e^2 & + & 4096, \\ a^2 ced & + & 27, \\ ab^2 ed^2 & + & 4670, \\ ac^4 & + & 18, \\ b^2 ce^2 & - & 9, \\ bcd^2 e & - & 4, \\ b^2 c^2 d & + & 1, \end{array}$$

we have

$$\Sigma \theta_i \theta_j = a^{-3} M e.$$

Next,

$$\Sigma \theta_1 \theta_2 \theta_3 = \Sigma \beta \gamma \delta \zeta^4(\beta, \gamma, \delta) \cdot -\gamma \delta \alpha \zeta^4(\gamma, \delta, \alpha) \cdot \delta \alpha \beta \zeta^4(\delta, \alpha, \beta) \quad [\text{four terms in all}],$$

giving, after collecting,

$$\begin{aligned} &= \alpha^2 \beta^2 \gamma^2 \delta^2 (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta) \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) \\ &= -a^{-6} e^2 \sqrt{\square} \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta), \end{aligned}$$

and observing that

$$\begin{aligned} (a, b, c, d, e \mid v, 1)^4 &= a(v - \alpha)(v - \beta)(v - \gamma)(v - \delta), \\ 4av^3 + 3bv^2 + 2cv + d &= a(v - \beta)(v - \gamma)(v - \delta) + \&c., \end{aligned}$$

which, putting $v = \alpha$, gives

$$4a\alpha^3 + 3b\alpha^2 + 2c\alpha + d = a(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta),$$

we have

$$\begin{aligned} \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) &= \frac{1}{a} (4a \Sigma \alpha^4 + 3b \Sigma \alpha^3 + 2c \Sigma \alpha^2 + d \Sigma \alpha) \\ &= 4(4) + 3b(3) + 2c(2) + d(1), \end{aligned}$$

where for a moment a is put equal to 1. This is calculated by the array

$$\begin{array}{rcll} e & -16 & & -16, \\ bd & +16 & -9 & -1 & +6, \\ c^2 & +8 & & -4 & +4, \\ b^2 c & -16 & +9 & +2 & -5, \\ b^4 & +4 & -3 & & +1, \end{array}$$

giving the collected terms displayed on the following page.

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or restoring the powers of a , and putting

$$\begin{array}{rcl} a^3e & -16, \\ a^2bd & +6, \\ a^2c^2 & +4, \\ ab^2c & -5, \\ b^4 & +1, \end{array}$$

we have

$$\Sigma_4\theta_1\theta_2\theta_3 = a^{-2} Ne\sqrt{\square}.$$

Lastly,

$$\begin{aligned} \theta_1\theta_2\theta_3\theta_4 \quad (\text{or } \theta_1\theta_2\theta_3\theta_4) &= \alpha^2\beta^2\gamma^2\delta^2 \zeta^4(\beta, \gamma, \delta) \zeta^4(\gamma, \delta, \alpha) \zeta^4(\delta, \alpha, \beta) \zeta^4(\alpha, \beta, \gamma) \\ &= \alpha^2\beta^2\gamma^2\delta^2 \zeta^4(\alpha, \beta, \gamma, \delta) \\ &= a^{-2} e^2 \square, \end{aligned}$$

and the equation $\Pi_1(\theta - \theta_i) = 0$ is thus found to be the one given above, No. 10.

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ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 225–239. Received February 20,—Read March 7, 1861.]

The Porism referred to is as follows, viz. two conics may be so related to each other, that a polygon may be inscribed in the one, and circumscribed about the other conic, in such manner that any point whatever of the circumscribing conic may be taken for a vertex of the polygon. I gave in the year 1853, in the *Philosophical Magazine*¹, a general formula for the relation between the two conics, viz. if $U = 0$ is the equation of the inscribed conic, $V = 0$ that of the circumscribed conic, and if $\text{disct.}(U + \xi V)$, where ξ is an arbitrary multiplier, denotes the discriminant of $U + \xi V$ in regard to the coordinates (x, y, z) (such discriminant being of course a cubic function of ξ , in regard to ξ , and also in regard to the coefficients of the two conics, U , V , jointly), then if we write

$$\sqrt{\text{disct.}(U + \xi V)} = A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + \&c.,$$

the relations for the cases of the triangle, pentagon, heptagon, &c. are

$$C = 0, \quad \begin{vmatrix} C & D \\ D & E \end{vmatrix} = 0, \quad \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix} = 0, \quad \&c.$$

¹See the papers — “On the Geometrical Representation of the Integral $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$,” *Phil. Mag.*, April 1853, [113].

“Note on the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* August 1853, [115].

“Correction of two Theorems relating to the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* November 1853, [116].

“Developments of the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* May 1854, [128].

respectively, while those in the cases of the quadrangle, hexagon, octagon, &c., are

$$D = 0, \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = 0, \quad \begin{vmatrix} D, & E, & F \\ E, & F, & G \\ F, & G, & H \end{vmatrix} = 0, \quad \&c.$$

respectively. The demonstration of this fundamental theorem is for greater completeness here reproduced; but the chief object of the memoir is to direct attention to a curious analytical theorem which is an easy *à priori* consequence of the Porism, and to obtain the relations for the several polygons up to the enneagon, in a new and simple form which puts in evidence *à posteriori* for these cases the analytical theorem just referred to. The analytical theorem rests upon the following considerations:—the relation for a hexagon ought to include that for a triangle; in fact a triangle with its sides in order twice over is a form of hexagon; the condition for an octagon should in like manner include that for a quadrangle; and so in other cases. Let the cubic function, $\text{disct.}(U + \xi V)$, be represented by $1 + \alpha\beta\xi + \gamma\xi^2 + \delta\xi^3$; the coefficients A, B, C, D, E , &c. are functions of β, γ, δ . Write

$$\begin{aligned} C &= (3), \quad D = (4), \\ \begin{vmatrix} C, & D \\ D, & E \end{vmatrix} &= (5), \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = (6), \\ \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix} &= (7), \end{aligned}$$

then (3), (4), (5) are respectively prime functions of β, γ, δ ; that is they cannot be decomposed into factors, rational functions of these quantities; and it is convenient to denote this by writing (3) = [3], (4) = [4], (5) = [5]. But by what precedes, (6) contains the factor (3), that is [3]; and if the other factor, which is prime, is denoted by [6], then we have (6) = [6][3]. The next term (7) is prime, that is we have (7) = [7]; but we have (8) = [8][4]; the term (9) gives (9) = [9][3], and so on. Thus we have (12) = [12][6][4][3], the numbers in () being all the factors, so the number itself included, and as well composite as prime, of the number in (), the factors 2 and 1 being however excluded. To make this clearer, it may be remarked that the last-mentioned equation has the geometrical signification that the relation for a dodecagon is the aggregate of the relations for a proper dodecagon, a proper hexagon, a quadrangle, and a triangle; that is, the relation for a dodecagon implies one or other of the last-mentioned relations. The relations for the several polygons up to the

enneagon are in the memoir obtained in a form which puts in evidence the property in question, that is, the series of equations

$$\begin{aligned}(3) &= [3], \\(4) &= [4], \\(5) &= [5], \\(6) &= [6][3], \\(7) &= [7], \\(8) &= [8][4], \\(9) &= [9][3].\end{aligned}$$

To do this, the discriminant is represented, not as above in terms of the constants β, γ, δ , but in a somewhat different form, by means of the constants $b [= \beta]$, c , d , the last two whereof are such that $c = 0$ is the relation for the triangle, $d = 0$ the relation for the quadrangle; thus $[3] = c$, $[4] = d$, and for the particular cases considered, the analytical theorem consists herein, that c is a factor of (6), and of (9), and that d is a factor of (8). I have, for the sake of homogeneity, introduced into the formulæ the quantity $a (= 1)$, but this is a matter of form only.

The functions $[3]$, $[4]$, &c. have been spoken of as prime; they are so, in fact, so far as they are calculated; and that they are so in general rests on the assumption that for a polygon of a given number of sides, there is but one form of relation: if, for instance, in the equation $[12] = 0$, which is the condition for a proper dodecagon, the function $[12]$ could be decomposed into rational factors, then equating each of these factors to zero, we should have so many distinct forms of relation for a proper dodecagon. I believe that the assumption and reasoning are valid; but without entering further into this, I take it for granted that in the general case the functions $[3]$, $[4]$, &c. are in fact prime. But the coefficients β, γ, δ , or b, c, d instead of being so many independent arbitrary quantities, may be given as rational functions of other quantities (if, for instance, the two conics are circles, radii R, r , and distance between the centres a , then β, γ, δ will be functions of R, r, a): and it is in a case of this kind quite conceivable that the functions $[3]$, $[4]$, &c., considered as functions of these new elements, should cease to be prime functions. In fact, in the case just referred to of the two circles (the original case of the Porism as considered by Fuss), the functions $[4]$, $[6]$, &c., which correspond to a polygon of an even number of sides, appear to be each of them decomposable into two factors: the memoir contains some remarks tending to show *à priori* that in the case in question this decomposition takes place. I was led to examine the point by the elegant formulæ obtained in an essentially different manner by M. Mention, *Bull. de l'Acad. de St Pétersbourg*, t. 1. pp. 15, 30, and 507 (1860), in reference to the case of the two circles (it thereby appears that the decomposition takes place for the quadrangle and the hexagon); and these formulæ are reproduced in the memoir.

Demonstration of the general Formula of the Relation between the two Conics.

The equation of a conic passing through the points of intersection of the conics

$$U = 0, \quad V = 0,$$

is of the form

$$U + mV = 0,$$

where m is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of m a quadratic equation; and conversely, if the roots of this quadratic equation are given, the line is also given; that is, the two roots may be considered as parameters which determine the particular line.

Let k be a given value of m ; the parameters of any tangent of the conic $U + kV = 0$ are k, p , but as k is common to all the tangents, we may consider the particular tangent as determined by the single parameter p . And a point of the conic $U + kV = 0$ may be considered as determined by the same parameter p which determines the tangent at that point.

As regards the conic $V = 0$, the common parameter for all its tangents is ∞ , and we may consider any other tangent of this conic as determined by the parameter θ , and a point of the conic as determined by the same parameter θ .

Suppose, in the first instance, that the two conics are

$$\begin{aligned} U &= ax^2 + by^2 + cz^2 = 0, \\ V &= x^2 + y^2 + z^2 = 0; \end{aligned}$$

the equation of the tangent of $U + kV = 0$, the parameter whereof is p (in fact a common tangent of the conics $U + kV = 0$, $U + pV = 0$), is easily found to be

$$x\sqrt{b-c}\sqrt{a+k}\sqrt{a+p} + y\sqrt{c-a}\sqrt{b+k}\sqrt{b+p} + z\sqrt{a-b}\sqrt{c+k}\sqrt{c+p} = 0;$$

and if this meet the conic $V = 0$ in the points P, P' , the parameters whereof are k, θ and k, θ' , or say θ and θ' respectively, then the coordinates of the point P are given by

$$x : y : z = \sqrt{b-c}\sqrt{a+\theta} : \sqrt{c-a}\sqrt{b+\theta} : \sqrt{a-b}\sqrt{c+\theta};$$

and substituting these values in the foregoing equation, we have

$$(b-c)\sqrt{a+k}\sqrt{a+p}\sqrt{a+\theta} + (c-a)\sqrt{b+k}\sqrt{b+p}\sqrt{b+\theta} + (a-b)\sqrt{c+k}\sqrt{c+p}\sqrt{c+\theta} = 0,$$

as an equation connecting the parameters p and θ . This equation may be replaced by

$$\begin{aligned} \sqrt{(a+k)(a+p)(a+\theta)} &= \lambda + \mu a, \\ \sqrt{(b+k)(b+p)(b+\theta)} &= \lambda + \mu b, \\ \sqrt{(c+k)(c+p)(c+\theta)} &= \lambda + \mu c. \end{aligned}$$

from which λ, μ are to be eliminated; and squaring and reducing, we have

$$\begin{aligned}\lambda^2 &= abc + kp\theta, \\ -2\lambda\mu &= bc + ca + ab - (p\theta + kp - k\theta),\end{aligned}$$

and thence

$$(bc + ca + ab - p\theta - kp - k\theta)^2 - 4(a + b + c + k + p + \theta)(abc + kp\theta) = 0$$

as the rational form of the original equation. But the same rational equation would, it is clear, be obtained from the system

$$\begin{aligned}\sqrt{(k+a)(k+b)(k+c)} &= L + Mk, \\ \sqrt{(p+a)(p+b)(p+c)} &= L + Mp, \\ \sqrt{(\theta+a)(\theta+b)(\theta+c)} &= L + M\theta,\end{aligned}$$

by the elimination of L and M . And it follows from Abel's theorem (but the result might be verified by means of Euler's fundamental theorem for the addition of elliptic functions), that if

$$\Pi\xi = \int \frac{d\xi}{\sqrt{(\xi+a)(\xi+b)(\xi+c)}},$$

then the last-mentioned system is equivalent to the transcendental equation

$$\Pi\theta = \Pi p \pm \Pi k,$$

in which the arbitrary constant which should have been inserted, and the sign of Πk , are determined by the consideration that for $k = \infty$ (which gives $\Pi k = 0$) we ought to have $\theta = p$, and therefore $\Pi\theta = \Pi p$.

There is of course a similar equation in θ' , and the terms with Πk must be taken with opposite signs, and we have thence the theorem:

"If θ, θ' are the parameters of the points P, P' in which the conic $V = 0$ is intersected by the tangent, the parameter whereof is p , of the conic $U + kV = 0$, then the equations

$$\begin{aligned}\Pi\theta &= \Pi p - \Pi k, \\ \Pi\theta' &= \Pi p + \Pi k,\end{aligned}$$

determine the parameters θ, θ' of the points in question." And again:

"If the two variable parameters θ, θ' are connected by the equation

$$\Pi\theta' - \Pi\theta = 2\Pi k,$$

then the line PP' will be a tangent of the conic $U + kV = 0$."

The foregoing demonstration relates to the particular forms $U = ax^2 + by^2 + cz^2$, $V = x^2 + y^2 + z^2$; but observing that the function $\sqrt{(\xi + a)(\xi + b)(\xi + c)}$, which enters under the integral sign in the transcendental function $\Pi\xi$, is the square root of the discriminant of $U + \xi V$, the theory of covariants shows at once that the conclusions apply to any forms whatever of U, V , the expression for the transcendental function being

$$\Pi\xi = \int \frac{d\xi}{\sqrt{\text{discr.}(U + \xi V)}}.$$

Consider now a triangle inscribed in the conic $V = 0$, and with its sides touching the conics

$$U + kV = 0, \quad U + k'V = 0, \quad U + k''V = 0,$$

then if $\theta, \theta', \theta''$ are the parameters of the angles, we have

$$\begin{aligned} \Pi\theta' - \Pi\theta &= 2\Pi k, \\ \Pi\theta'' - \Pi\theta' &= 2\Pi k', \\ \Pi\theta - \Pi\theta'' &= 2\Pi k'', \end{aligned}$$

and thence

$$\Pi k + \Pi k' + \Pi k'' = 0,$$

as the relation which must subsist between the parameters k, k', k'' of the conics touched by the sides; and similarly for a polygon of n sides, the relation between the parameters is

$$\Pi k_1 + \Pi k_2 + \cdots + \Pi k_n = 0.$$

But by Abel's theorem, this transcendental equation is equivalent to an algebraical one.

In fact, calling the radical $\sqrt{D\xi}$, then if $\varphi x, \chi x$ are rational and integral functions of x with arbitrary coefficients, and if

$$\varphi^2 x - y^2 \chi^2 x \cdot Dx = A(x - k_1)(x - k_2) \cdots (x - k_n),$$

(this implies that φx is of a degree not exceeding n and χx of a degree not exceeding $n - 3$; that is, for n even the degrees of φ, χ are $\frac{1}{2}n, \frac{1}{2}(n - 4)$; and for n odd they are $\frac{1}{2}(n - 1), \frac{1}{2}(n - 3)$), then the algebraical equation is that obtained by the elimination of the arbitrary coefficients from the system of equations

$$\begin{aligned} \varphi k_1 + \chi k_1 \sqrt{Dk_1} &= 0, \\ \varphi k_2 + \chi k_2 \sqrt{Dk_2} &= 0, \\ &\vdots \\ \varphi k_n + \chi k_n \sqrt{Dk_n} &= 0; \end{aligned}$$

or, what is the same thing, for n odd, $= 2p - 1$, it is

$$\{1, \theta, \dots, \theta^{p-1}, \sqrt{D\theta}, \theta\sqrt{D\theta}, \dots, \theta^{p-2}\sqrt{D\theta}\} = 0,$$

and for n even, $= 2p$, it is

$$\{1, \theta, \dots, \theta^p, \sqrt{D\theta}, \theta\sqrt{D\theta}, \dots, \theta^{p-2}\sqrt{D\theta}\} = 0,$$

where the expressions in $\{ \}$ denote respectively the determinants, of $2p - 1$ lines, and $2p$ lines, formed by substituting for θ the values $k_1, k_2, \dots, k_n$ respectively. Thus for $n = 3$, the equation is

$$\begin{vmatrix} 1, & k_1, & \sqrt{Dk_1} \\ 1, & k_2, & \sqrt{Dk_2} \\ 1, & k_3, & \sqrt{Dk_3} \end{vmatrix} = 0;$$

and for $n = 4$, it is

$$\begin{vmatrix} 1, & k_1, & k_1^2, & \sqrt{Dk_1} \\ 1, & k_2, & k_2^2, & \sqrt{Dk_2} \\ 1, & k_3, & k_3^2, & \sqrt{Dk_3} \\ 1, & k_4, & k_4^2, & \sqrt{Dk_4} \end{vmatrix} = 0,$$

and so on.

Suppose

$$\sqrt{D\xi} = A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \dots;$$

then substituting the corresponding expressions for $\sqrt{Dk_1}, \sqrt{Dk_2}$, &c., the determinant will divide by $\{1, \theta, \theta^2, \dots, \theta^{p-1}\}$, and it may be seen without difficulty that the resulting equation on putting therein $k_1 = k_2 = \dots = k_n = 0$, will, according as $n = 3, 4, 5, 6$, &c., &c., be

$$C = 0, \quad D = 0, \quad \begin{vmatrix} C, & D \\ D, & E \end{vmatrix}, \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix}, \quad \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix},$$

which is the theorem above referred to.

II.

Application to the several Polygons up to the Enneagon.

If the equation of the inscribed conic is $U = 0$, and that of the circumscribed conic is $V = 0$, and if the discriminant $(U + \xi V)$ is in the first instance represented by $1 + 4\beta\xi + 4\gamma\xi^2 + 4\delta\xi^3$, then the square root of the discriminant is

$$1 + 2\beta\xi + 2(\gamma - \beta^2)\xi^2 + 2(\delta - 2\beta\gamma + 2\beta^3)\xi^3 + \&c.,$$

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so that the condition for the triangle is

$$\gamma - \beta^2 = 0,$$

and that for the quadrangle is

$$\delta - 2\beta\gamma + 2\beta^3 = 0.$$

It is obviously convenient to introduce into the formulæ, in the place of γ and δ , the quantities

$$c = \gamma - \beta^2, \quad d = \delta - 2\beta\gamma + 2\beta^3;$$

and writing also, for symmetry of notation, b in the place of β , we have

$$\beta = b, \quad \gamma = c + b^2, \quad \delta = d + 2bc,$$

so that the discriminant will be

$$= 1 + 4b\xi + 4(c + b^2)\xi^2 + 4(d + 2bc)\xi^3,$$

which is

$$= (1 + 2b\xi + 2c\xi^2)^2 + 4(d\xi^3 - c^2\xi^4).$$

But for homogeneity I introduce the quantity $a = 1$, and put the discriminant

$$= (1 + 2b\xi + 2ac\xi^2)^2 + 4a^2(d\xi^3 - c^2\xi^4).$$

The square root, divided by a^2 , is

$$a^{-2}(1 + 2b\xi + 2ac\xi^2) \sqrt{1 + \{4a^2(d\xi^3 - c^2\xi^4)\} \div (1 + 2b\xi + 2ac\xi^2)^2};$$

or developing, this is

$$\begin{aligned} & a^{-2}(1 + 2b\xi + 2ac\xi^2) \\ & + 2(d\xi^3 - c^2\xi^4) \div (1 + 2b\xi + 2ac\xi^2) \\ & - 2a^2(d\xi^3 - c^2\xi^4)^2 \div (1 + 2b\xi + 2ac\xi^2)^3 \\ & + 4a^4(d\xi^3 - c^2\xi^4)^3 \div (1 + 2b\xi + 2ac\xi^2)^5 \\ & - 10a^6(d\xi^3 - c^2\xi^4)^4 \div (1 + 2b\xi + 2ac\xi^2)^7 \\ & + \&c.; \end{aligned}$$

or representing this by $1 + 2B\xi + 2C\xi^2 + 2D\xi^3 + \&c.$, we have

$$C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + H\xi^7 + I\xi^8 + J\xi^9 + K\xi^{10} + L\xi^{11} + M\xi^{12} + \dots$$

[the expansion of the foregoing is a large array of monomial terms in a, b, c, d with signed integer coefficients, occupying the body of the page; representative entries include

$$\begin{aligned} & (1 - 2b\xi - 2ac\xi^2) \text{ contributes the leading terms,} \\ & \text{further terms include } -8b^3\xi^3, -24ab^2c\xi^4, +64a^3bc^3\xi^6, \\ & \text{up to } +16a^4c\xi^8 - 160a^4bc^4\xi^9 + \&c.], \end{aligned}$$

and the values of the coefficients C, D, E , &c. thus are:

the following table sets out the coefficients $C, D, E, F, G, H, I, J, K, L, M$ as polynomials in a, b, c, d (with $a = 1$):

$C :$	$-1 \cdot c,$	leading term;
$D :$	$+1 \cdot d,$	
$E :$	$bd - 2, \quad c^2 - 1,$	
$F :$	$6abcd + 2, \quad ac^3 + 2,$	
$G :$	$a^2d^2 - 1, \quad a^2bd^2 + 6, \quad a^2bcd + 8, \quad ab^3d^2 - 24,$ $ab^2c^2 - 8, \quad b^4d - 32, \quad \&c.$	
$H :$	$a^3bd^2 + 6, \quad a^3cd^2 + 6, \quad a^3b^2cd^2 - 245,$ $a^2b^4cd - 360, \quad ab^4d + 168, \quad b^6 - 32, \quad \&c.$	
$I :$	$a^4d^3 - 1, \quad a^4bd^3 + 2, \quad a^3b^2cd^2 - 1320,$ $a^3c^2d^2 - 1320, \dots, \quad a^2b^4d^2 + 240, \quad a^2c^4d + 30,$ $b^4cd - 160, \quad b^6d - 64, \quad \&c.$	
$J :$	$a^5bd^3 + 2, \quad a^4b^2cd^2 + 144, \quad a^4b^3d^3 + 360,$ $a^4bcd^3 + 1480, \quad a^3b^2c^2d^2 - 98000,$ $a^3b^3cd^2 - 672240, \quad a^3b^5d^2 + 560, \quad \&c.$	
$K :$	$a^5b^2d^2 - 300, \quad a^5bc^2d^2 - 560, \quad a^4b^3cd^2 + 160,$ $a^4bc^3d + 11600, \quad \&c.$	
$L :$	$a^5c^4 - 5, \quad a^5bc^3d + 1400, \quad \&c.$	
$M :$	$\&c.;$	

But in the sequel only the coefficients up to I are made use of: the expressions for J, K, L, M may however be useful, and they are given accordingly.

The sums of the numerical coefficients are given here and elsewhere, as they are very useful for verifications; thus, putting $a = b = c = d = 1$, we have, as should be,

$$\sqrt{1 + 4\xi + 8\xi^2 + 12\xi^3} = 1 + 2\xi + 2\xi^2 (1, +1, -3, +4, -3, +4, -19, +60, -124, +214, -455, \&c.)(1, \xi)^n.$$

Proceeding now to form the several terms of the matrix

$$(C, D, E, F, G, \dots),$$

$$(D, E, F, G, H, \dots),$$

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&c.,

which may be represented by 12, 13, &c., viz.

$$12 = \begin{vmatrix} C & D \\ D & E \end{vmatrix}, \quad 13 = \begin{vmatrix} C & E \\ D & F \end{vmatrix},$$

we have, up to 45, which is all that is required:

$$\begin{aligned} 12 &= a^{-1} \times (ad^2 - 1, bcd - 2, c^3 - 1), \\ 13 &= a^{-1} \times (abd^2 + 2, ac^2d - 1, b^2cd + 4, bc^3 + 2), \\ 14 &= a^{-1} \times (a^2cd^2 + 1, ab^2d^2 - 4, abc^2d + 6, ac^4 + 2, b^3cd - 8, b^2c^3 - 4), \\ 15 &= a^{-1} \times (a^3d^3 + 1, a^2bcd^2 - 2, a^2c^2d + 4, ab^3d^2 + 8, ab^2c^2d - 20, abc^4 - 8, b^4cd + 16, b^3c^3 + 8), \\ 23 &= (acd^2 - 2, bc^2d - 2, c^4 - 1), \\ 24 &= (a^2d^3 - 1, abcd^2 + 4, b^2c^2d + 4, bc^4 + 2), \\ 25 &= (a^2bd^3 + 4, a^2c^2d^2 + 5, ab^2cd^2 - 8, abc^3d + 4, ac^5 + 2, b^3c^2d - 8, b^2c^4 - 4), \\ 34 &= (a^2bd^3 + 2, a^2c^2d^2 - 3, ab^2cd^2 - 4, ac^5 - 2), \\ 35 &= (a^3cd^3 - 2, a^2b^2d^3 - 8, a^2c^3d - 2, ab^3cd^2 + 8, abc^5 + 4), \\ 45 &= (a^4d^4 - 1, a^4bcd^3 + 4, a^3c^2d^2 - 8, a^3b^3d^3 + 8, a^3b^2c^2d^2 + 12, a^2bc^4d - 4, a^2c^6 - 4). \end{aligned}$$

Forming in like manner the determinants of the matrix

$$(C, D, E, F, \dots),$$

$$(D, E, F, G, \dots),$$

$$(E, F, G, H, \dots),$$

and representing these by 123, 124, &c., viz.

$$123 = \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix}, \text{ \&c.},$$

we have, up to 234,

$$\begin{aligned} 123 &= a^{-2} \times (a^3d^4 + 1, a^2bcd^3 + 2, a^2c^3d^2 - 1, abcd^4 - 2, ac^6 - 1), \\ 124 &= a^{-1} \times (a^2bd^4 - 4, a^2c^2d^3 - 3, a^2b^2cd^3 - 8, a^2bc^3d^2 - 4, a^2c^5d - 2, ab^3c^2d^2 + 4, abc^6 + 2), \\ 134 &= a^{-1} \times (a^4cd^4 + 1, a^3b^2d^4 + 4, a^3bc^2d^3 + 8, a^3c^4d^2 - 3, a^2b^3cd^3 + 8, a^2b^2c^3d^2 + 12, a^2c^7 - 2), \\ 234 &= (a^4d^5 - 1, a^3bcd^4 - 4, a^3c^3d^3 - 4, a^2b^2c^2d^3 - 4, a^2bc^4d^2 - 6, a^2c^6d - 2). \end{aligned}$$

and further, the determinants of the matrix

$$\begin{aligned} & (C, D, E, F, \dots), \\ & (D, E, F, G, \dots), \\ & (E, F, G, H, \dots), \\ & (F, G, H, I, \dots), \end{aligned}$$

in the present case, the single determinant

$$1234 = \begin{vmatrix} C, & D, & E, & F \\ D, & E, & F, & G \\ E, & F, & G, & H \\ F, & G, & H, & I \end{vmatrix},$$

we have

$$1234 = a^{-1} \times (a^6 cd^6 - 3, a^5 bc^2 d^5 - 12, a^5 c^4 d^4 - 4, a^4 b^2 c^3 d^4 - 16, a^4 bc^5 d^3 - 14, a^4 c^7 d^2 - 3, a^3 b^3 c^4 d^3 - 8, a^3 b^2 c^6 d^2 - 12, \dots)$$

The conditions for the triangle and the quadrangle are $c = 0, d = 0$; those for the pentagon and hexagon are $12 = 0, 23 = 0$; for the heptagon and octagon, $123 = 0, 234 = 0$; and that for the enneagon is $1234 = 0$. The foregoing values show that 23 and 1234 (which belong to the hexagon and the enneagon) divide by c (which belongs to the triangle), and that 234 (which belongs to the octagon) divides by d (which belongs to the quadrangle). But I was not prepared for the destruction which will be observed in the several determinants, of the terms involving the lower powers of a (that is, the terms of the highest orders in b, c, d), and which renders these expressions so much more simple than they would otherwise have been.

Representing the reduced equations for the several polygons, as before, by

$$\begin{aligned} [3] &= 0, \\ [4] &= 0, \\ [5] &= 0, \\ [6] &= 0, \\ [7] &= 0, \\ [8] &= 0, \\ [9] &= 0, \text{ \&c.}, \end{aligned}$$

then retaining the quantity $a (= 1)$ for homogeneity, but rejecting the powers of a which divide out, and reversing in some cases the signs, the values of the functions [3], [4], &c. are

$$\begin{aligned} [3] &= c + 1, \\ [4] &= d + 1, \\ [5] &= ad^2 + 1, \quad bcd + 2, \quad c^3 + 1, \\ [6] &= ad^2 + 2, \quad bcd + 2, \quad c^3 + 1, \\ [7] &= a^2d^4 + 1, \quad abcd^3 + 2, \quad ac^3d^2 - 1, \quad bc^4d - 2, \quad c^6 - 1, \\ [8] &= a^2d^4 + 1, \quad abcd^3 + 4, \quad ac^3d^2 + 4, \quad b^2cd^2 + 4, \quad bc^4d + 6, \quad c^6 + 2, \\ [9] &= a^3cd^6 + 3, \quad a^2bc^2d^5 + 12, \quad a^2c^4d^4 + 4, \quad ab^2c^3d^4 + 16, \quad abc^5d^3 - 14, \quad ac^7d^2 + 3, \quad b^3c^4d^3 + 8, \quad b^2c^6d^2 + 4. \end{aligned}$$

The similarity of form for the relations corresponding to the pentagon and the hexagon, and for those corresponding to the heptagon and the octagon, is, I am inclined to think, accidental; the functions are homogeneous as regards degree and weight; and the degrees and weights of the two consecutive functions being identical, the literal parts must be similarly constituted.

III.

M. Mention's Formulæ for the Case of two Circles.

In the case of two circles, if, as usual, the radii of the inscribed and circumscribed circles are put equal to r and R respectively, and the distance of their centres to a , then the equations of the inscribed and circumscribed circles respectively may be taken to be

$$\begin{aligned} x^2 + y^2 - r^2 &= 0, \\ (x - a)^2 + y^2 - R^2 &= 0; \end{aligned}$$

and if, in the notation of M. Mention, we put

$$\frac{R^2 - r^2}{r^2} = \nu, \quad \frac{1}{r^4}(r^4 + R^4 + a^4 - 2r^2R^2 - 2r^2a^2 - 2R^2a^2) = -\nu',$$

then the quadratic radical is

$$\sqrt{(1 + 4\nu\xi)\{[1 + (2 + 2\nu)\xi]^2 + 4\nu'\xi^3\}}.$$